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Aim:Key Exchange using Diffie-Hellman

Implement the Diffie-Hellman key exchange algorithm to securely exchange keys between two entities over an insecure network.

Code:CLI

```
import random

def generate_private_key():
    return random.randint(2, 100)

def generate_public_key(private_key, g, p):
    return pow(g, private_key, p)

def generate_shared_key(public_key, private_key, p):
    return pow(public_key, private_key, p)

def main():
    print("DIFFIE-HELLMAN KEY EXCHANGE")
    print("=" * 40)

    p = int(input("Enter prime number p: "))
    g = int(input("Enter primitive root g: "))

    alice_private = generate_private_key()
    bob_private = generate_private_key()

    alice_public = generate_public_key(
        alice_private,
        g,
        p
    )

    bob_public = generate_public_key(
        bob_private,
        g,
        p
    )
```

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```
)

alice_shared = generate_shared_key(
    bob_public,
    alice_private,
    p
)

bob_shared = generate_shared_key(
    alice_public,
    bob_private,
    p
)

print("\nGazala Private Key:", alice_private)
print("Gazala Public Key:", alice_public)

print("\nShreya Private Key:", bob_private)
print("Shreya Public Key:", bob_public)

print("\nGazala Shared Key:", alice_shared)
print("Shreya Shared Key:", bob_shared)

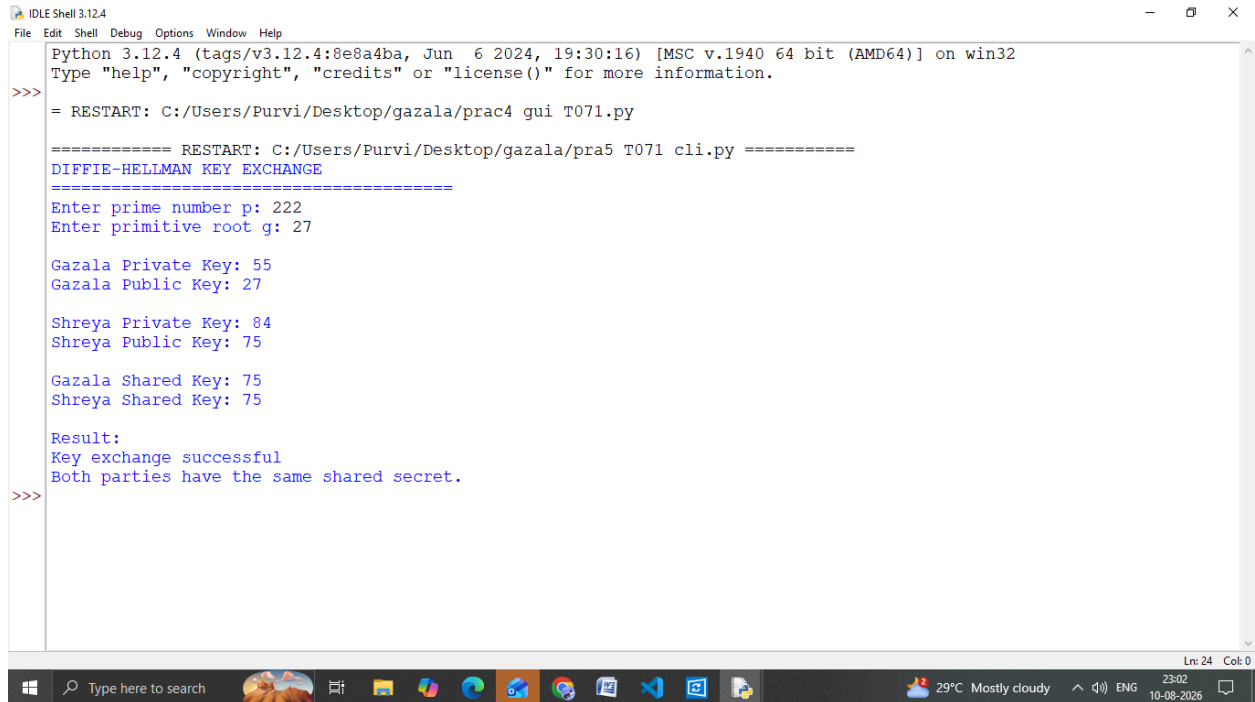
print("\nResult:")

if alice_shared == bob_shared:
    print("Key exchange successful")
    print("Both parties have the same shared secret.")
else:
    print("Key exchange failed")

if __name__ == "__main__":
    main()
```

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Output:



```
Python 3.12.4 (tags/v3.12.4:8e8a4ba, Jun 6 2024, 19:30:16) [MSC v.1940 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/Purvi/Desktop/gazala/prac4 gui T071.py

===== RESTART: C:/Users/Purvi/Desktop/gazala/prac5 T071 cli.py =====
DIFFIE-HELLMAN KEY EXCHANGE
=====
Enter prime number p: 222
Enter primitive root g: 27

Gazala Private Key: 55
Gazala Public Key: 27

Shreya Private Key: 84
Shreya Public Key: 75

Gazala Shared Key: 75
Shreya Shared Key: 75

Result:
Key exchange successful
Both parties have the same shared secret.
>>>
```

Code:GUI

```
import tkinter as tk
from tkinter import messagebox
import random

def private_key():
    return random.randint(2, 10)

def public_key(private, g, p):
    return pow(g, private, p)

def shared_key(public, private, p):
    return pow(public, private, p)

def exchange():
    try:
        p = int(p_entry.get())
        g = int(g_entry.get())

        private1 = private_key()
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```

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```
private2 = private_key()

public1 = public_key(private1, g, p)
public2 = public_key(private2, g, p)

shared1 = shared_key(public2, private1, p)
shared2 = shared_key(public1, private2, p)

private1_entry.delete(0, tk.END)
private1_entry.insert(0, private1)

public1_entry.delete(0, tk.END)
public1_entry.insert(0, public1)

private2_entry.delete(0, tk.END)
private2_entry.insert(0, private2)

public2_entry.delete(0, tk.END)
public2_entry.insert(0, public2)

shared1_entry.delete(0, tk.END)
shared1_entry.insert(0, shared1)

shared2_entry.delete(0, tk.END)
shared2_entry.insert(0, shared2)

if shared1 == shared2:
    result.config(
        text="KEY EXCHANGE SUCCESSFUL\nGazala and Shreya have the same secret
key.",
        fg="green"
    )
else:
    result.config(
        text="KEY EXCHANGE FAILED",
        fg="red"
    )

except ValueError:
    messagebox.showerror("Error", "Please enter valid numbers")
```

```
root = tk.Tk()
root.title("Diffie-Hellman Key Exchange")
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```

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```
root.geometry("600x650")
root.config(bg="lightgray")
```

```
tk.Label(
    root,
    text="DIFFIE-HELLMAN KEY EXCHANGE",
    font=("Arial", 20, "bold"),
    bg="lightgray"
).pack(pady=15)
```

```
tk.Label(
    root,
    text="Prime Number (p)",
    bg="lightgray"
).pack()
```

```
p_entry = tk.Entry(root)
p_entry.pack()
p_entry.insert(0, "23")
```

```
tk.Label(
    root,
    text="Primitive Root (g)",
    bg="lightgray"
).pack()
```

```
g_entry = tk.Entry(root)
g_entry.pack()
g_entry.insert(0, "5")
```

```
person1_label = tk.Label(
    root,
    text="Gazala",
    font=("Arial", 14, "bold"),
    bg="lightgray"
)
person1_label.pack(pady=10)
```

```
private1_entry = tk.Entry(root, width=40)
private1_entry.pack()
```

```
tk.Label(
    root,
    text="Gazala Private Key",
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```

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```
        bg="lightgray"
    ).pack()

    public1_entry = tk.Entry(root, width=40)
    public1_entry.pack()

    tk.Label(
        root,
        text="Gazala Public Key",
        bg="lightgray"
    ).pack()

    person2_label = tk.Label(
        root,
        text="Shreya",
        font=("Arial", 14, "bold"),
        bg="lightgray"
    )
    person2_label.pack(pady=10)

    private2_entry = tk.Entry(root, width=40)
    private2_entry.pack()

    tk.Label(
        root,
        text="Shreya Private Key",
        bg="lightgray"
    ).pack()

    public2_entry = tk.Entry(root, width=40)
    public2_entry.pack()

    tk.Label(
        root,
        text="Shreya Public Key",
        bg="lightgray"
    ).pack()

    shared1_entry = tk.Entry(root, width=40)
    shared1_entry.pack(pady=5)

    tk.Label(
        root,
        text="Gazala Shared Key",
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```

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```
        bg="lightgray"
    ).pack()

    shared2_entry = tk.Entry(root, width=40)
    shared2_entry.pack(pady=5)

    tk.Label(
        root,
        text="Shreya Shared Key",
        bg="lightgray"
    ).pack()

    tk.Button(
        root,
        text="Perform Key Exchange",
        command=exchange,
        bg="blue",
        fg="white",
        font=("Arial", 11, "bold")
    ).pack(pady=15)

    result = tk.Label(
        root,
        text="Click the button to perform key exchange",
        font=("Arial", 11, "bold"),
        bg="lightgray",
        fg="blue"
    )
    result.pack()
    root.mainloop()
```

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Output:

Diffie-Hellman Key Exchange

DIFFIE-HELLMAN KEY EXCHANGE

Prime Number (p)
84

Primitive Root (g)
5

Gazala

10
Gazala Private Key

1
Gazala Public Key

Shreya

3
Shreya Private Key

37
Shreya Public Key

1
Gazala Shared Key

1
Shreya Shared Key

Perform Key Exchange

KEY EXCHANGE SUCCESSFUL
Gazala and Shreya have the same secret key.

Windows taskbar: Type here to search, 29°C Mostly cloudy, ENG, 23:05, 10-08-2026

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